



Computer Networks Lab

CL-3001 | Fall 2026 | Section BAI-5A

Course Instructor: Mr. Abdullah Shaikh

LAB # 05 — TASKS

Introduction to FTP, SMTP and their Implementation

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Section	BAI-5A	Date of Submission	

Submission Guidelines

- Complete all tasks and paste the screenshots of every configuration and every test into this Word file.
- Submit the completed file as a **PDF** on GCR, named **LabCourseName_Lab#-YourName YourRollNo/ID** — e.g. **CN_Lab#05-Abdullah Shaikh-25K-7826**.
- Upload all **.pkt** and **.pcap** files to your Google Drive and grant access to abdullah.shaikh@nu.edu.pk.
- The router in every topology must be configured through the CLI, not the GUI.

Label every device on the workspace with its IP and MAC address using **Place**

Note. • *No deadline will be extended under any circumstances.*

OBJECTIVE

To understand how the File Transfer Protocol (FTP) and the Simple Mail Transfer Protocol (SMTP) work at the application layer, to implement FTP, and SMTP services on a routed network in Cisco Packet Tracer.

TASK # 01 — FTP SERVER ON A ROUTED NETWORK

Design and implement the topology given in Figure 1 in Cisco Packet Tracer. The network consists of two subnets joined by a router: a client LAN and a server LAN that hosts the FTP server. The router must be configured through the CLI. Attach all the necessary screenshots in this Word file and upload the PKT file to your drive.

Addressing Requirement

Derive both network addresses from the last four digits of your roll number:

- Client LAN (Fa0/0 side): **XX.XX.0.0 / 16** — e.g. roll number 4129 → **41.29.0.0 / 16**

- Server LAN (Fa0/1 side): **XX.0.0.0 / 8** — e.g. roll number 4129 → **29.0.0.0 / 8**

Every host must be given a static IP address, the correct subnet mask and the correct default gateway.
Complete Table 1 before you begin configuring the devices.

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Fa0/0	10.13.0.1	255.255.0.0	—
Router0	Fa0/1	13.0.0.1	255.0.0.0	—
PC0	Fa0	10.13.0.10	255.255.0.0	10.13.0.1
PC1	Fa0	10.13.0.11	255.255.0.0	10.13.0.1
PC2	Fa0	10.13.0.12	255.255.0.0	10.13.0.1
PC3	Fa0	13.0.0.10	255.0.0.0	13.0.0.1
FTP Server	Fa0	13.0.0.2	255.0.0.0	13.0.0.1

Table 1: IP addressing table for Task # 01

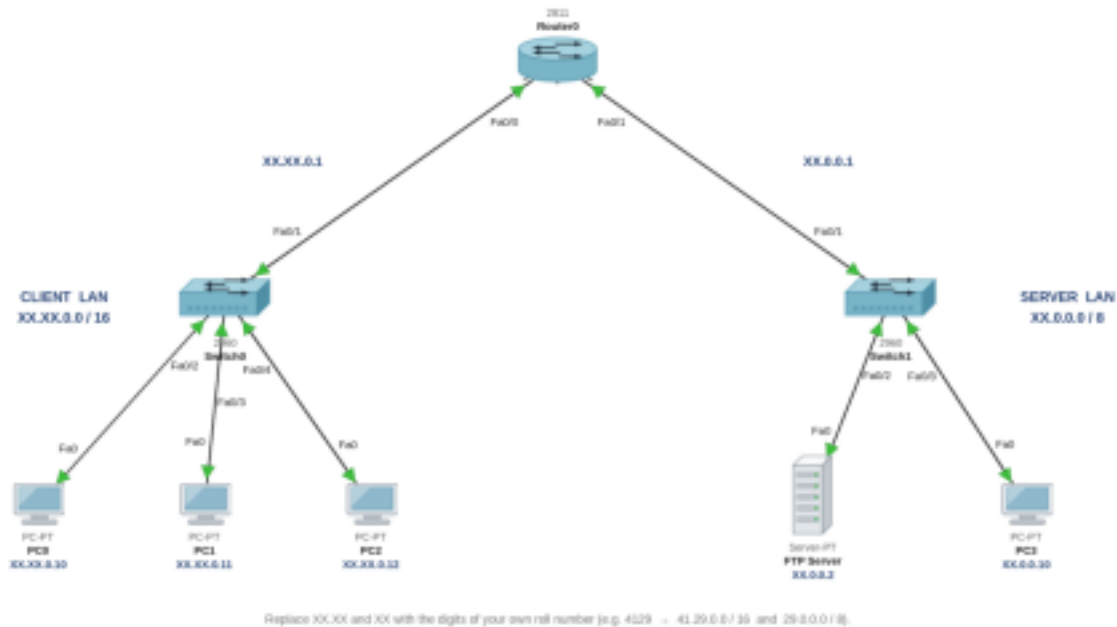


Figure 1: Topology for Task # 01

Implementation Steps

1. Build the topology shown in Figure 1 using one 2811 router, two 2960 switches, four PCs and one Server-PT. Connect all devices with straight-through copper cables.

2. Configure both router interfaces through the CLI (replace the placeholders with your own addresses):
3. Assign the static addresses from Table 1 to every PC and to the server (Desktop → IP Configuration). Verify inter-subnet connectivity by pinging the FTP server from PC0 and from PC2.

4. On the server, open **Services** → **FTP**. Turn the service **On**, then create a user account in which the user name is your first name and the password is your roll number. Grant the Write, Read, Delete, Rename and List permissions, click **Add**, and remove the default **cisco** account.
5. On PC0 open **Desktop** → **Text Editor**, type your name, roll number and section, and save the file as **<YourRollNo>.txt**.
6. Upload the file from PC0 using the Command Prompt
7. Download the same file from a PC on the other subnet (PC3), then delete one file from the server and list the directory again:
8. Label every device on the workspace with its IP and MAC address using **Place Note**, and save the file as **CN_Lab05_Task01_<YourRollNo>.pkt**.

Screenshots Required

- Output of show ip interface brief and the running configuration of the router.
- Successful ping from a client on the client LAN to the FTP server.
- The FTP service page showing your user account and its permissions.
- The complete FTP session on both subnets, showing the login message, dir, put, get and delete outputs.

PASTE YOUR CLI CONFIGURATION, FTP SERVER SETUP AND CLIENT TEST SCREENSHOTS HERE

```
Router>enable
Router#configure terminal
      ^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet 0/0
Router(config-if)#ip address 10.13.0.1 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet 0/1
Router(config-if)#ip addresss 13.0.0.1 255.0.0.0
      ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 13.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#
```

Copy

Paste

Pinging 13.0.0.2 with 32 bytes of data:

```
Reply from 13.0.0.2: bytes=32 time<1ms TTL=127
Reply from 13.0.0.2: bytes=32 time<1ms TTL=127
Reply from 13.0.0.2: bytes=32 time<1ms TTL=127
Reply from 13.0.0.2: bytes=32 time<1ms TTL=127
```

Ping statistics for 13.0.0.2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

C:\>

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP
- PRP

FTP

Service ☒ On ☐ Off

User Setup

Username Password

☐ Write ☐ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission
1	Abdullah	0013	RWDNL

Add

Save

Remove

File

- 1 asa842-k8.bin
- 2 asa923-k8.bin
- 3 c1841-advipservicesk9-mz.124-15.T1.bin
- 4 c1841-ipbase-mz.123-14.T7.bin
- 5 c1841-ipbasek9-mz.124-12.bin
- 6 c1900-universalk9-mz.SPA.155-3.M4a.bin
- 7 c2600-advipservicesk9-mz.124-15.T1.bin
- 8 c2600-i-mz.122-28.bin

Remove

☐ Top

```
C:\>ftp 13.0.0.2
Trying to connect...13.0.0.2
Connected to 13.0.0.2
220- Welcome to PT Ftp server
Username:abdullah

%Error ftp://13.0.0.2/ (No such Account)
332- Need account for login

ftp 13.0.0.2
Trying to connect...13.0.0.2
Connected to 13.0.0.2
220- Welcome to PT Ftp server
Username:Abdullah
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>put 0013.txt

Writing file 0013.txt to 13.0.0.2:
File transfer in progress...

[Transfer complete - 31 bytes]

31 bytes copied in 0.044 secs (704 bytes/sec)
ftp>quit

221- Service closing control connection.
C:\>|
```

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 13.0.0.2
Trying to connect...13.0.0.2
Connected to 13.0.0.2
220- Welcome to PT Ftp server
Username:Abdullah
331- Username ok, need password
Password:****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 13.0.0.2:
 0  : 0013.txt                               31
 1  : asa842-k8.bin                          5571584
 2  : asa923-k8.bin                          30468096
 3  : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 4  : c1841-ipbase-mz.123-14.T7.bin          13832032
 5  : c1841-ipbasek9-mz.124-12.bin           16599160
 6  : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 7  : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 8  : c2600-i-mz.122-28.bin                  5571584
 9  : c2600-ipbasek9-mz.124-8.bin            13169700
10  : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
11  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
12  : c2800nm-ipbase-mz.123-14.T7.bin        5571584
13  : c2800nm-ipbasek9-mz.124-8.bin          15522644
14  : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
15  : c2950-i6q4l2-mz.121-22.EA4.bin        3058048
16  : c2950-i6q4l2-mz.121-22.EA8.bin        3117390
17  : c2960-lanbase-mz.122-25.FX.bin         4414921
18  : c2960-lanbase-mz.122-25.SEE1.bin       4670455
19  : c2960-lanbasek9-mz.150-2.SE4.bin       4670455
20  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
21  : c3560-advipservicesk9-mz.122-46.SE.bin 10713279
22  : c800-universalk9-mz.SPA.152-4.M4.bin    33591768
23  : c800-universalk9-mz.SPA.154-3.M6a.bin   83029236
24  : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
25  : cgr1000-universalk9-mz.SPA.154-2.CG     159487552
26  : cgr1000-universalk9-mz.SPA.156-3.CG     184530138
27  : ie9k_iosxe.17.09.04.SPA.bin           596133776
28  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
29  : ir800-universalk9-mz.SPA.155-3.M       61750062
30  : ir800-universalk9-mz.SPA.156-3.M       63753767
31  : ir800_yocto-1.7.2.tar                  2877440
32  : ir800_yocto-1.7.2_python-2.7.3.tar    6912000
```

```

ftp>get 0013.txt

Reading file 0013.txt from 13.0.0.2:
File transfer in progress...

[Transfer complete - 31 bytes]

31 bytes copied in 0.011 secs (2818 bytes/sec)
ftp>delete 0013.txt

Deleting file 0013.txt from 13.0.0.2: ftp>
[Deleted file 0013.txt successfully ]
ftp>dir

Listing /ftp directory from 13.0.0.2:
0   : asa842-k8.bin                5571584
1   : asa923-k8.bin                30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin  33591768
3   : c1841-ipbase-mz.123-14.T7.bin  13832032
4   : c1841-ipbasek9-mz.124-12.bin  16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin  33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin  33591768
7   : c2600-i-mz.122-28.bin        5571584
8   : c2600-ipbasek9-mz.124-8.bin   13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin  50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin  33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin  15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin  33591768
14  : c2950-i6q4l2-mz.121-22.EA4.bin  3058048
15  : c2950-i6q4l2-mz.121-22.EA8.bin  3117390
16  : c2960-lanbase-mz.122-25.FX.bin  4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin  4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin  4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin  8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin  10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin  83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin  505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG  159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG  184530138
26  : ie9k_iosxe.17.09.04.SPA.bin  596133776
27  : ir800-universalk9-bundle.SPA.156-3.M.bin  160968869

```

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TASK # 02 — SMTP AND POP3 EMAIL SERVICE

Design and implement the topology given in Figure 2. The mail server is connected directly to the router with a copper cross-over cable and provides three services: SMTP, POP3 and DNS. The clients must send and receive email using a domain name, not an IP address. The router must again be configured through the CLI.

Use the same roll-number-based addressing scheme as Task # 01: the client LAN is XX.XX.0.0 / 16 and the server network is XX.0.0.0 / 8.

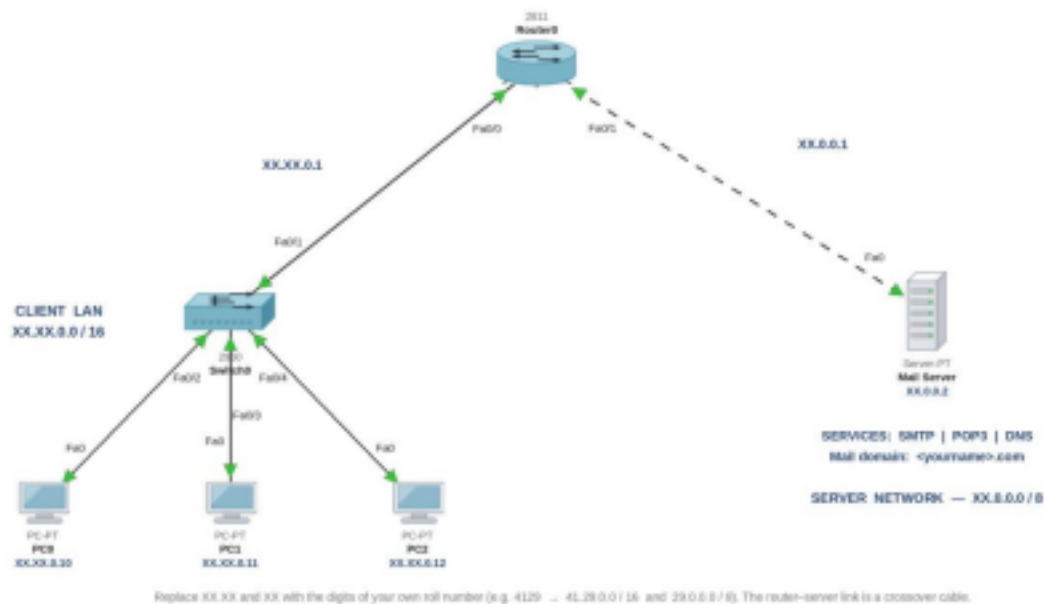


Figure 2: Topology for Task # 02

Implementation Steps

1. Build the topology of Figure 2. Use a copper cross-over cable between Router0 Fa0/1 and the mail server, and straight-through cables everywhere else.
2. Configure the router interfaces through the CLI exactly as in Task # 01, using Fa0/0 for the client LAN and Fa0/1 for the server network.
3. Assign static addresses to all hosts. On every PC, set the **DNS Server** field to the address of the mail server (XX.0.0.2).
4. On the server, open **Services** → **EMAIL**. Turn **SMTP** and **POP3** On, set the domain name to **<yourname>.com** and click Set. Then create one user account for each PC (user name and password of your choice) and click + to add each of them.
5. Open **Services** → **DNS**, turn the service On and add an A record that maps **mail.<yourname>.com** to the server address XX.0.0.2, then click Add.
6. On each PC, open **Desktop** → **E Mail** and configure the client: your name, the email address **<user>@<yourname>.com**, incoming and outgoing mail server **mail.<yourname>.com**, and the user name and password you created on the server. Click Save.

7. From PC0, compose and send an email to PC1 with the subject **CL-3001 Lab 05 – <Your Name> – <Your Roll No>**. On PC1 click Receive, open the message and reply to it. Confirm that PC0 receives the reply.
8. Send a second email from PC1 to PC2 to prove that all three accounts work, then attempt to send an email to a user that does not exist on the server and record the error message that the client reports.
9. Label all devices with their IP and MAC addresses using **Place Note** and save the file as **CN_Lab05_Task02_<YourRollNo>.pkt**.

Email Account Plan

Complete Table 2 with the accounts you created on the mail server.

Device	Display Name	Email Address	Incoming / Outgoing Server	Password
PC4	PC4 User	pc4@abdullah.com	mail.abdullah.com	pass0013
PC5	PC5 User	pc5@abdullah.com	mail.abdullah.com	pass0013
PC6	PC6 User	pc6@abdullah.com	mail.abdullah.com	pass0013

Table 2: Email accounts configured for Task # 02

Screenshots Required

- The EMAIL service page showing the domain name and the list of user accounts.
- The DNS service page showing the A record for the mail server.
- The mail client configuration of at least two PCs.
- The sent message, the received message on the destination PC, the reply, and the error message produced for the unknown recipient.

PASTE YOUR SERVER CONFIGURATION, MAIL CLIENT SETUP AND EMAIL TEST SCREENSHOTS HERE

Press RETURN to get started!

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 24.13.0.1 255.255.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 13.0.0.1 255.0.0.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Copy

Paste

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type **A Record** ▼

Address

Add **Save** **Remove**

No.	Name	Type	Detail
0	mail.abdullah.com	A Record	13.0.0.2

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP

EMAIL

SMTP Service ☒ ON ☐ OFF

POP3 Service ☒ ON ☐ OFF

Domain Name: **Set**

User Setup

User Password

pc4
pc5
pc6

MAIL BROWSER

X

Mails

Compose

Reply

Receive

Delete

Configure Mail

From

Subject

Received

Sending mail to pc5@abdullah.com , with subject : cn lab 5(ABDULLAH) .. Mail Server:
mail.abdullah.com
DNS resolving. Resolving name: mail.abdullah.com by querying to DNS Server: 13.0.0.2 DNS
resolved ip address: 13.0.0.2
Send Success.

Cancel
Send/Receive☐ Top

Message

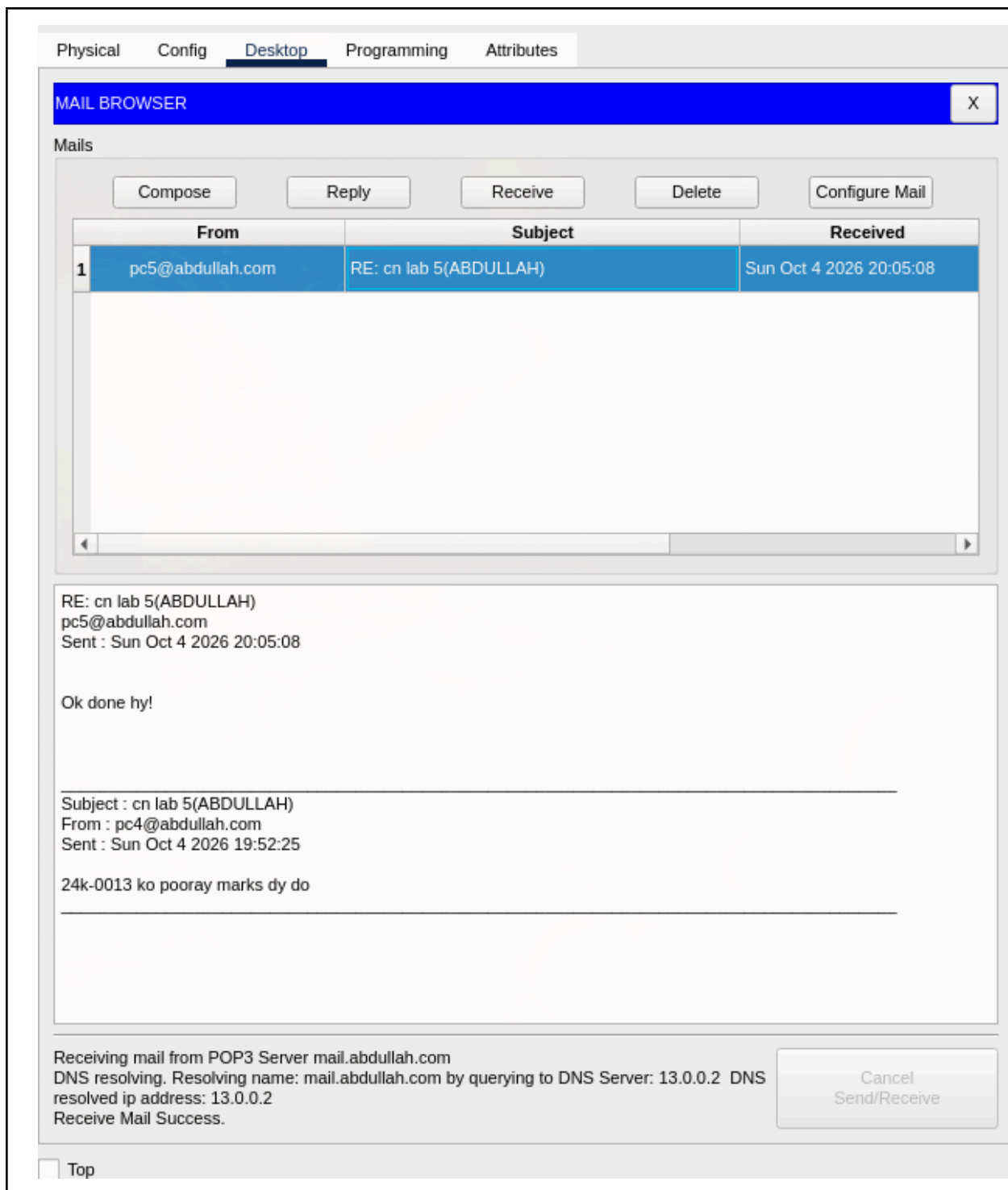
X

Reply

From: pc4@abdullah.com
To: pc5@abdullah.com
Subject: cn lab 5(ABDULLAH)

Sent: Sun Oct 4 2026 19:52:25

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The End!

Good luck — remember to upload your PKT files along with this document.